

PRDA – 03

HR ANALYTICS

PROJECT REPORT

Power BI HR Analytics Dashboard

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Tool Used: Microsoft Power BI

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Chapter 1 – Project Overview

This PRDA – 03 HR Analytics project analyzes employee workforce data and presents the results through a Power BI dashboard. The objective is to understand workforce size, demographics, work location, hiring and termination trends, departmental distribution, and other workforce characteristics.

The project brief requires the data to be obtained from the database, key HR KPIs to be created, required charts to be developed, the available features to be analyzed, and business insights and suggestions to be provided.

Chapter 2 – Project Objectives

1. Obtain the HR dataset and prepare it for analysis.
2. Clean and transform the data before calculations and visualization.
3. Create KPIs for total employees, remote employees, HQ employees, and average age.
4. Analyze hiring trends by year and gender distribution.
5. Analyze employee distribution by state, department, and race.
6. Analyze additional features such as age group, employee status, city, job title, and department-wise status.
7. Build the required Power BI visualizations.
8. Derive evidence-based insights and provide practical company suggestions.

Chapter 3 – Dataset Description

The PRDA – 03 project brief states that the dataset contains 22,214 records and employee information covering demographic, employment, location, hiring, and termination fields.

Field	Description
id	Unique employee identification number.
first_name / last_name	Employee name fields.
birthdate	Employee date of birth.
Age	Employee age.
gender	Employee gender.
race	Employee race category.
department	Employee department.
jobtitle	Employee job title.
location	Work location such as Headquarters or Remote.
hire_date	Employee hiring date.
termdate	Employee termination date, where recorded.
location_city	Employee work city.
location_state	Employee work state.

Chapter 4 – Data Connection and Preparation

4.1 Connecting the Data

The HR data was brought into Power BI using Excel as the source. Transform Data / Power Query was then used to prepare the dataset before analysis.

- Why: to bring the data into the Power BI environment and prepare it before calculations and visuals.

4.2 Cleaning termdate

The termdate field contained a date together with a time portion. Text Before Delimiter was used with a space delimiter to remove the time portion.

- Why: the project requires termination-date analysis, so the date portion is sufficient and easier to use consistently.

4.3 Converting termdate to Date

After removing the time portion, termdate was converted to the Date data type. Missing termination dates were kept blank/null.

- Why: Date type supports date-based analysis, while blank termination dates can be used to distinguish employees without a recorded termination date.

4.4 Close & Apply

Close & Apply was selected after the Power Query transformations.

- Why: this loads the prepared data into the Power BI model so DAX measures and visualizations can be created.

Chapter 5 – KPI Analysis

The following KPI values were taken directly from the completed Power BI dashboard.

KPI	Value	Meaning
Total Employees	22,214	Total employee records analyzed.
Average Age	38.43 years	Average employee age.
Remote Employee %	24.75%	Share classified as Remote.
HQ Employee %	75.25%	Share classified as Headquarters.

5.1 DAX Measures

Total Employees

Total Employees = COUNTROWS(Hr)

Remote Employees %

Remote Employees % = DIVIDE(CALCULATE(COUNTROWS(Hr), Hr[location] = "Remote"), [Total Employees], 0)

HQ Employees %

HQ Employees % = DIVIDE(CALCULATE(COUNTROWS(Hr), Hr[location] = "HQ"), [Total Employees], 0)

Average Age

Average Age = AVERAGE(Hr[Age])

5.2 KPI Interpretation

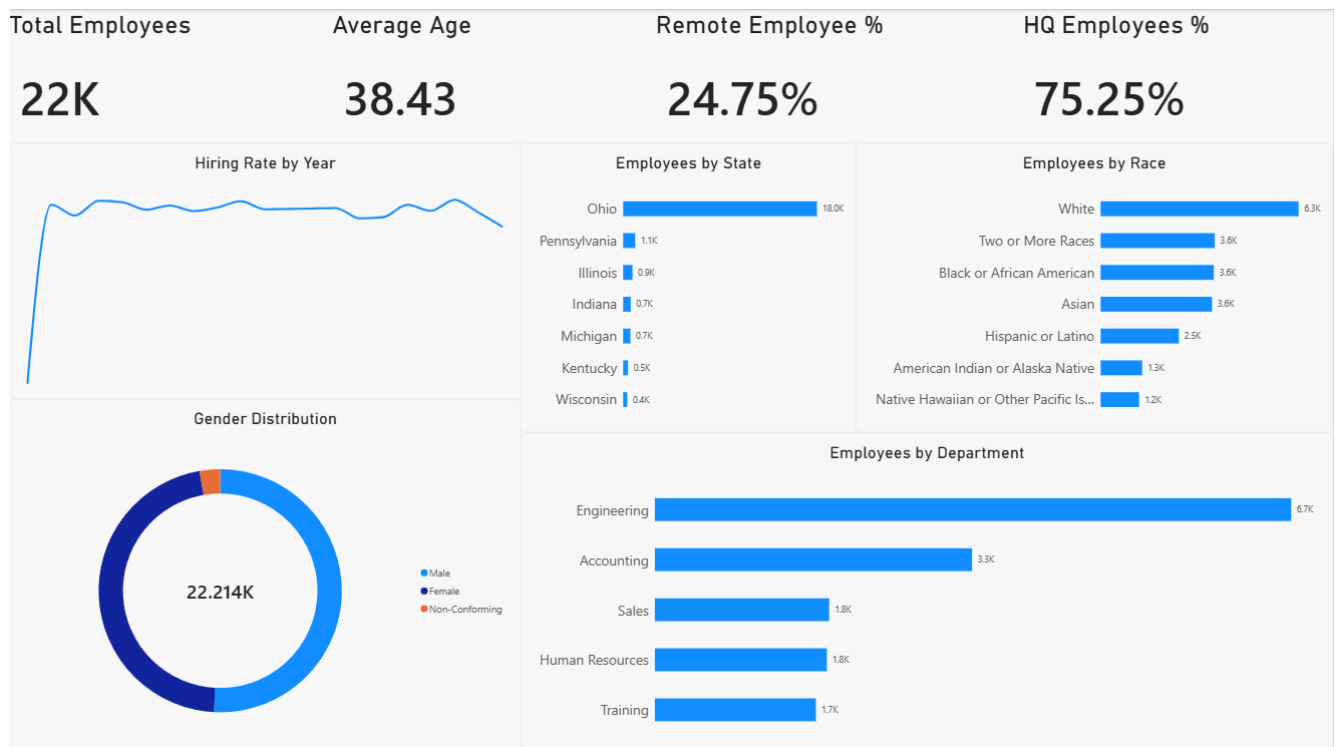
- The analysis covers 22,214 employee records.
- The average employee age is 38.43 years.
- 24.75% of employees are classified as Remote and 75.25% as Headquarters.
- The location KPIs show that the majority of the analyzed workforce is based at HQ.

Chapter 6 – Power BI Visualizations

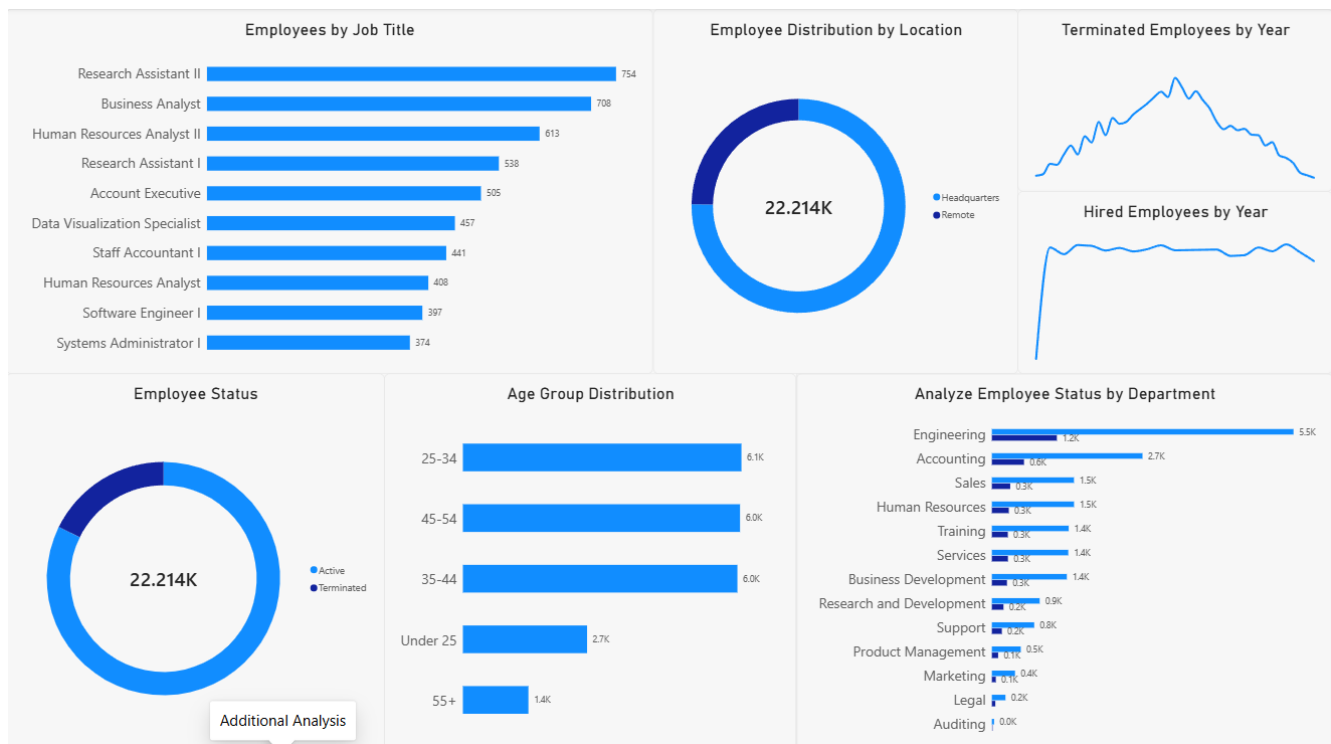
The dashboard contains the required project charts plus a limited number of additional visuals used to analyze important workforce features.

Visualization	Type	Purpose
Hired Employees by Year	Line chart	Shows hiring counts across years.
Gender Distribution	Donut chart	Shows employee distribution by gender.
Employees by State	Bar chart	Shows workforce concentration by state.
Employees by Department	Bar chart	Shows workforce size by department.
Employees by Race	Bar chart	Shows workforce distribution by race.
Employees by Job Title	Bar chart	Shows employee counts for displayed job titles.
Employee Distribution by Location	Donut chart	Compares Headquarters and Remote employees.
Terminated Employees by Year	Line chart	Shows termination counts across years.
Employee Status	Donut chart	Compares Active and Terminated employees.
Age Group Distribution	Bar chart	Shows employee distribution by age group.
Employee Status by Department	Stacked bar chart	Compares Active and Terminated counts across departments.

Terminology note: The visual originally titled “Hiring Rate by Year” is based on a hired-employee count measure. It is therefore described here as “Hired Employees by Year,” unless a separate hiring-rate formula is implemented.



Dashboard 1



Dashboard 2

Shows hiring counts across years.



Fig 1

Shows employee distribution by gender.

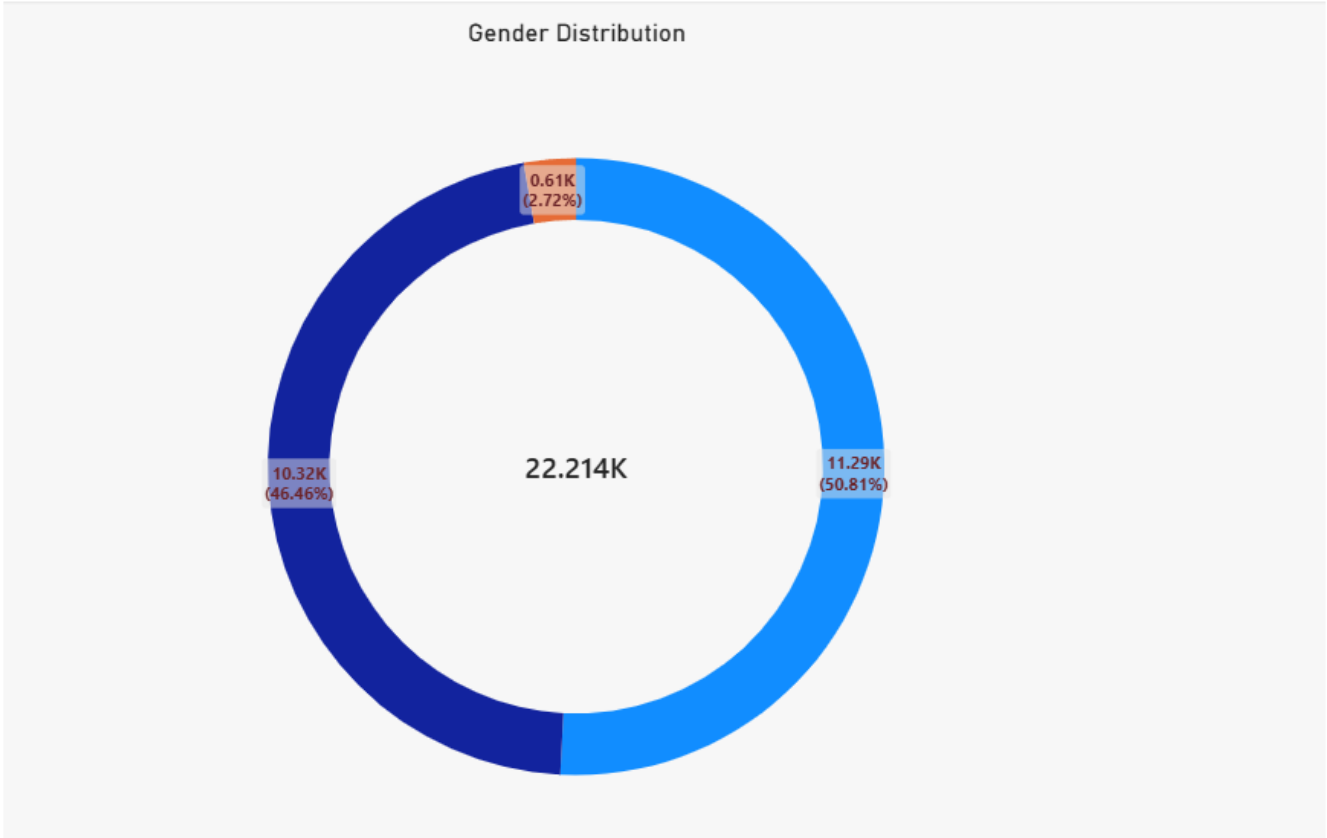


Fig 2

Shows workforce concentration by state.

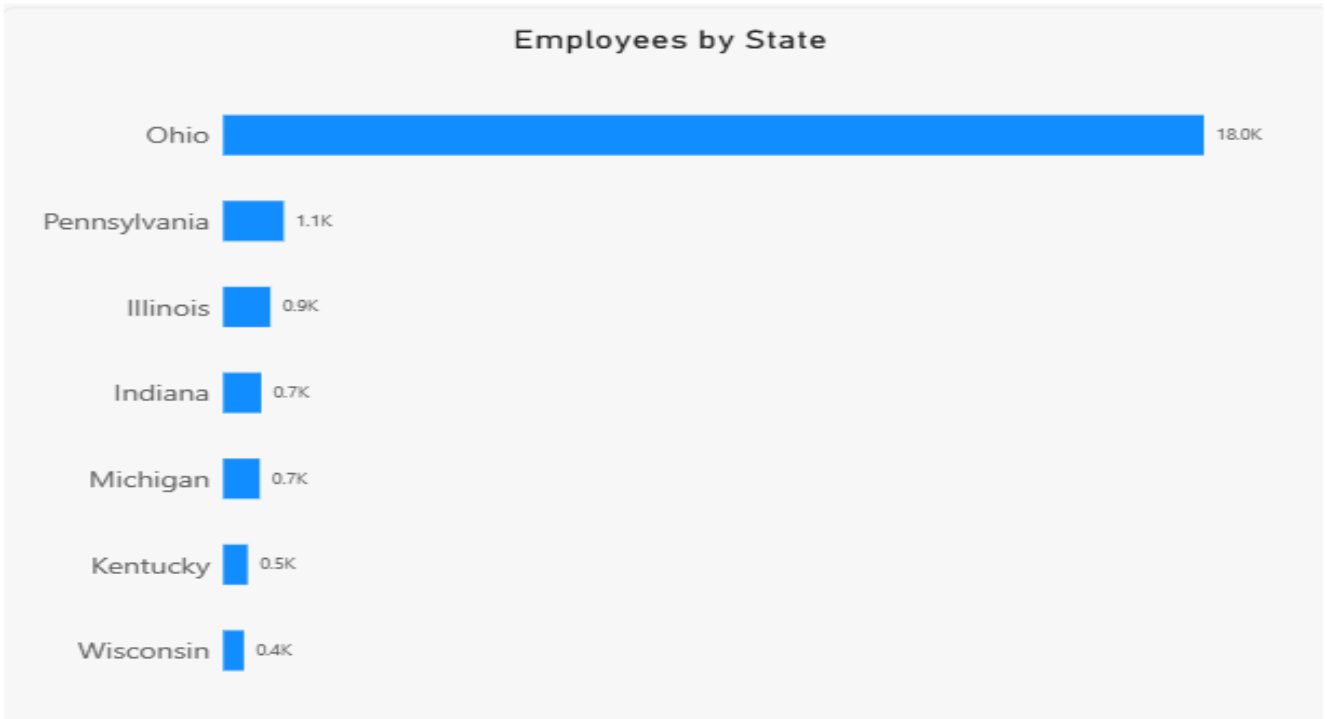


Fig 3

Shows workforce size by department.

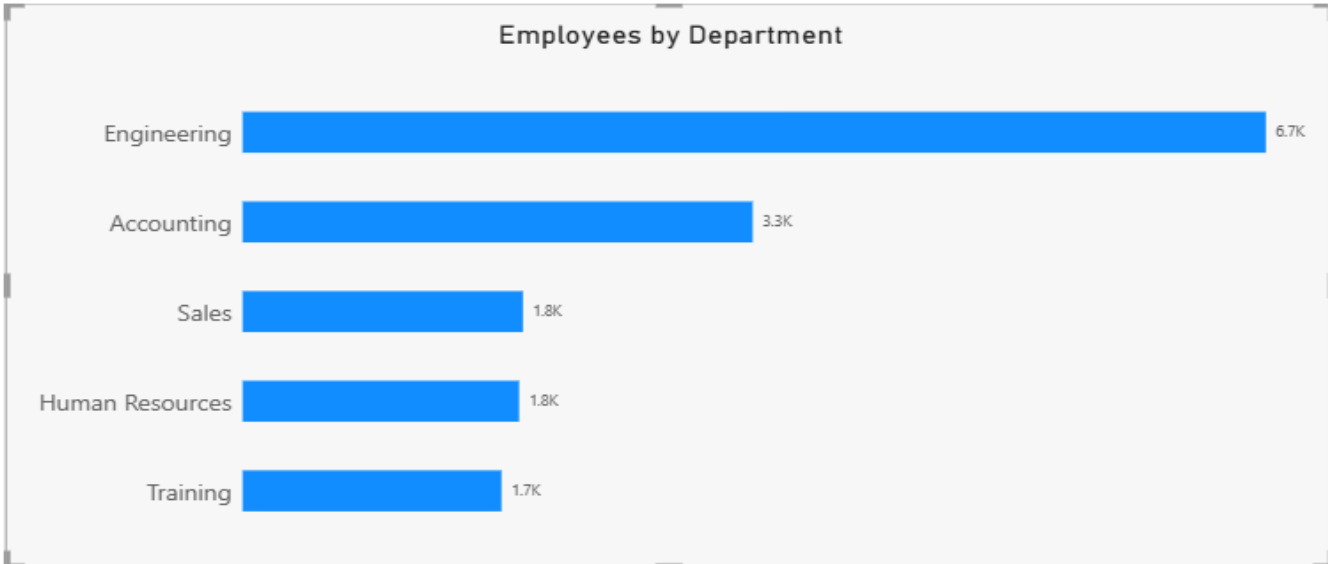


Fig 4

Shows workforce distribution by race.

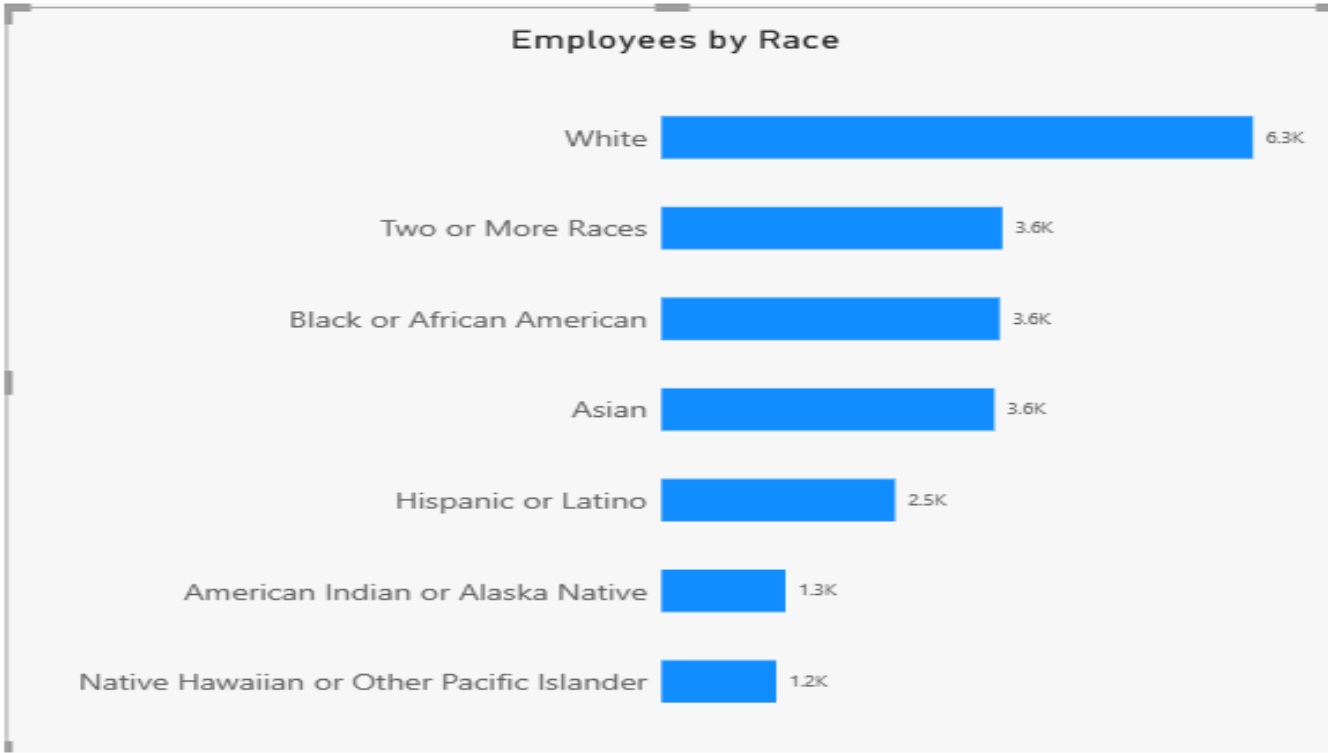


Fig 5

Shows TOP 10 employee counts for displayed job titles.

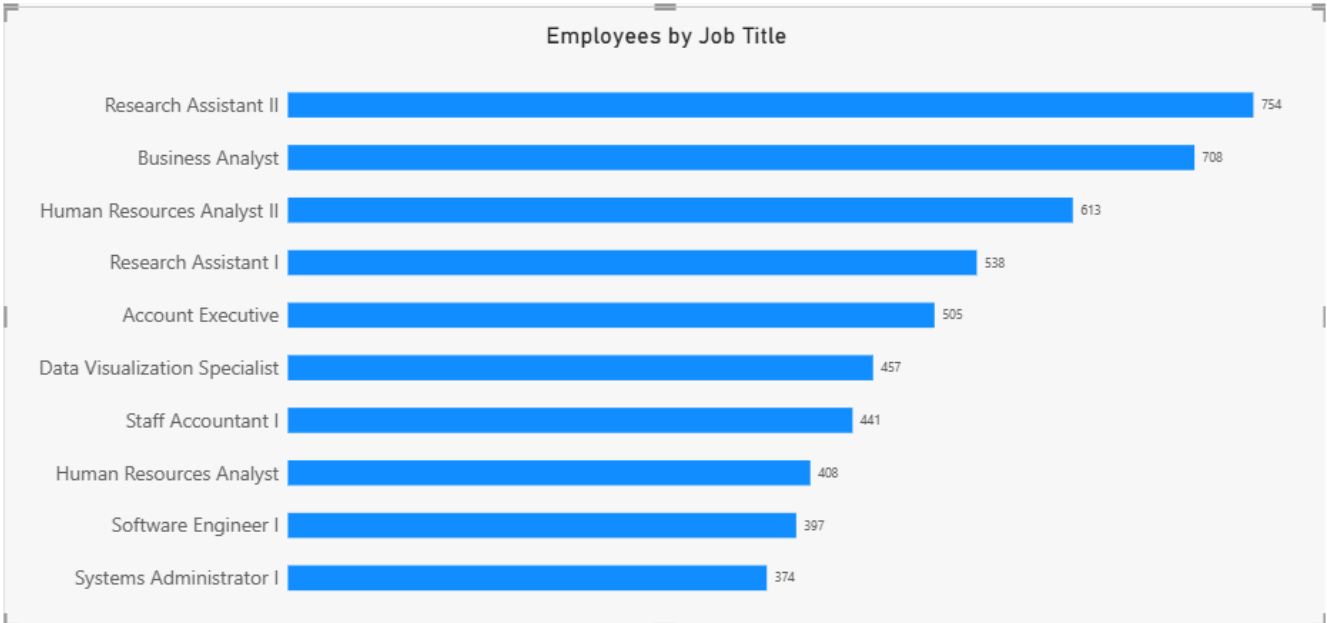


Fig 6

Compares Headquarters and Remote employees.

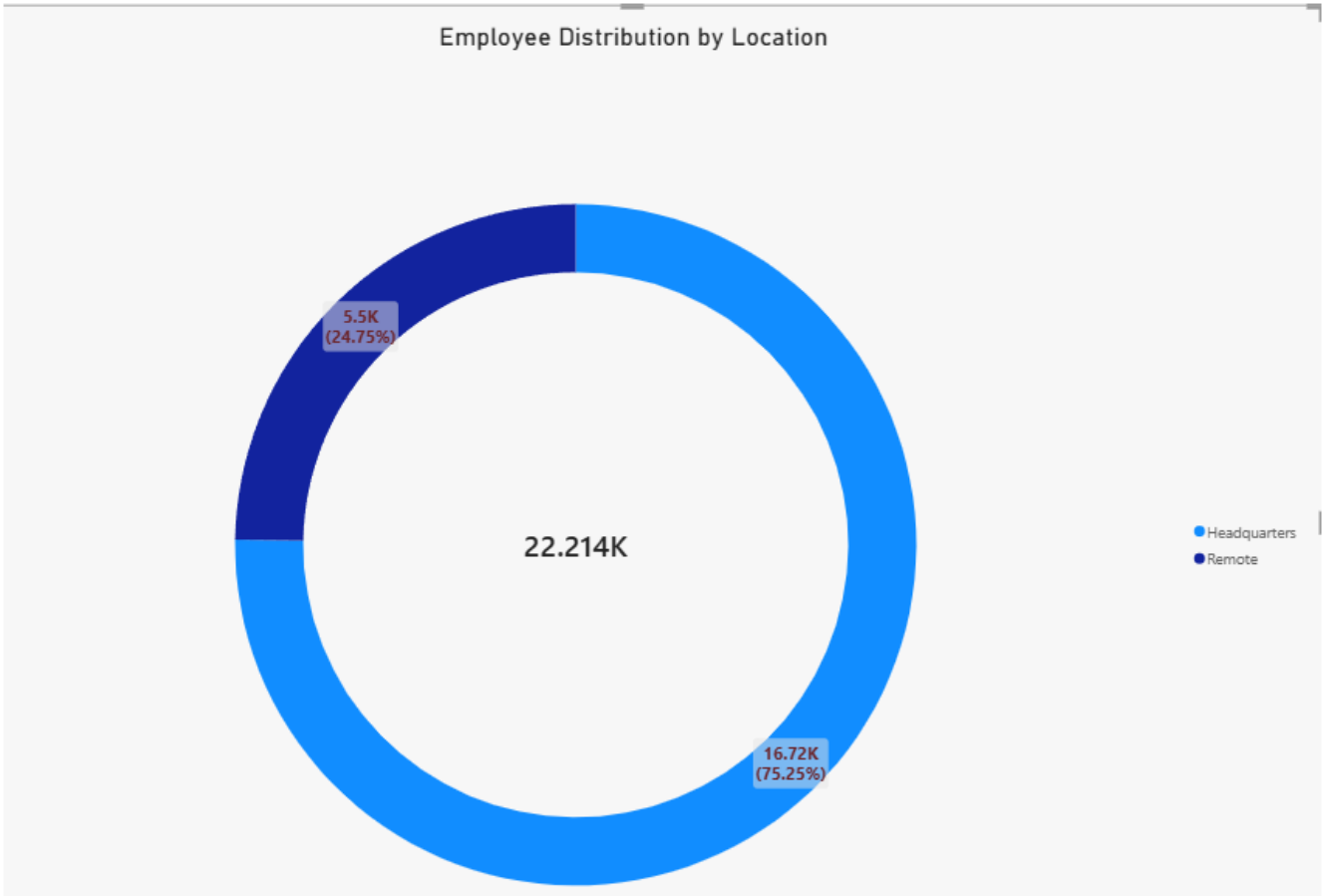


Fig 7

Shows termination counts across years.

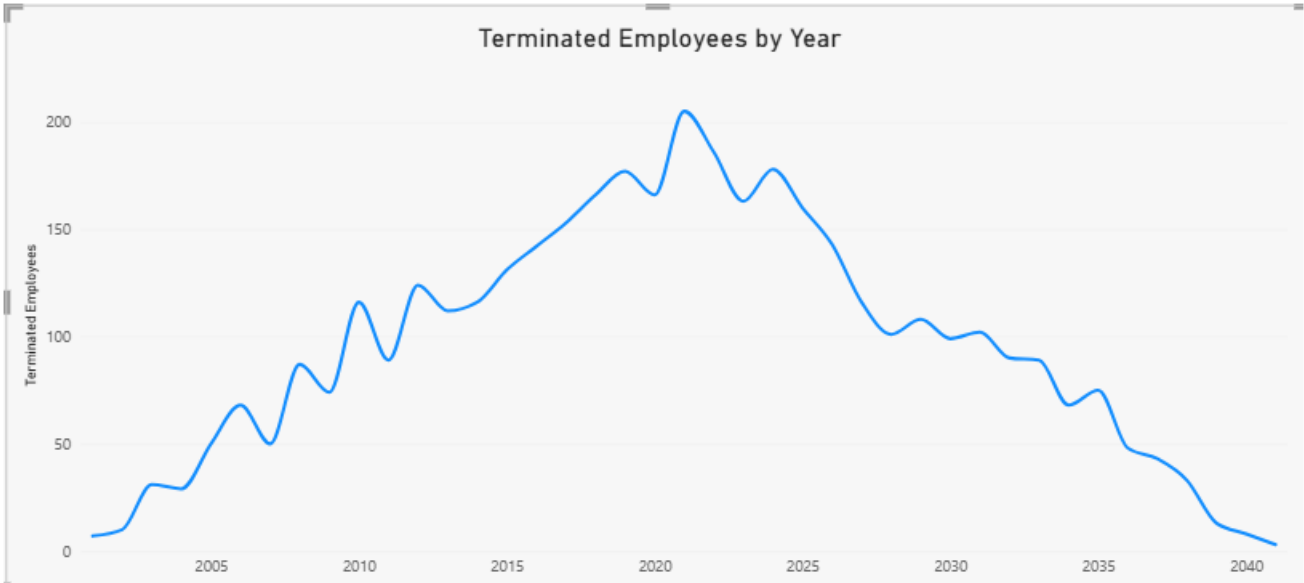


Fig 8

Compares Active and Terminated employees.

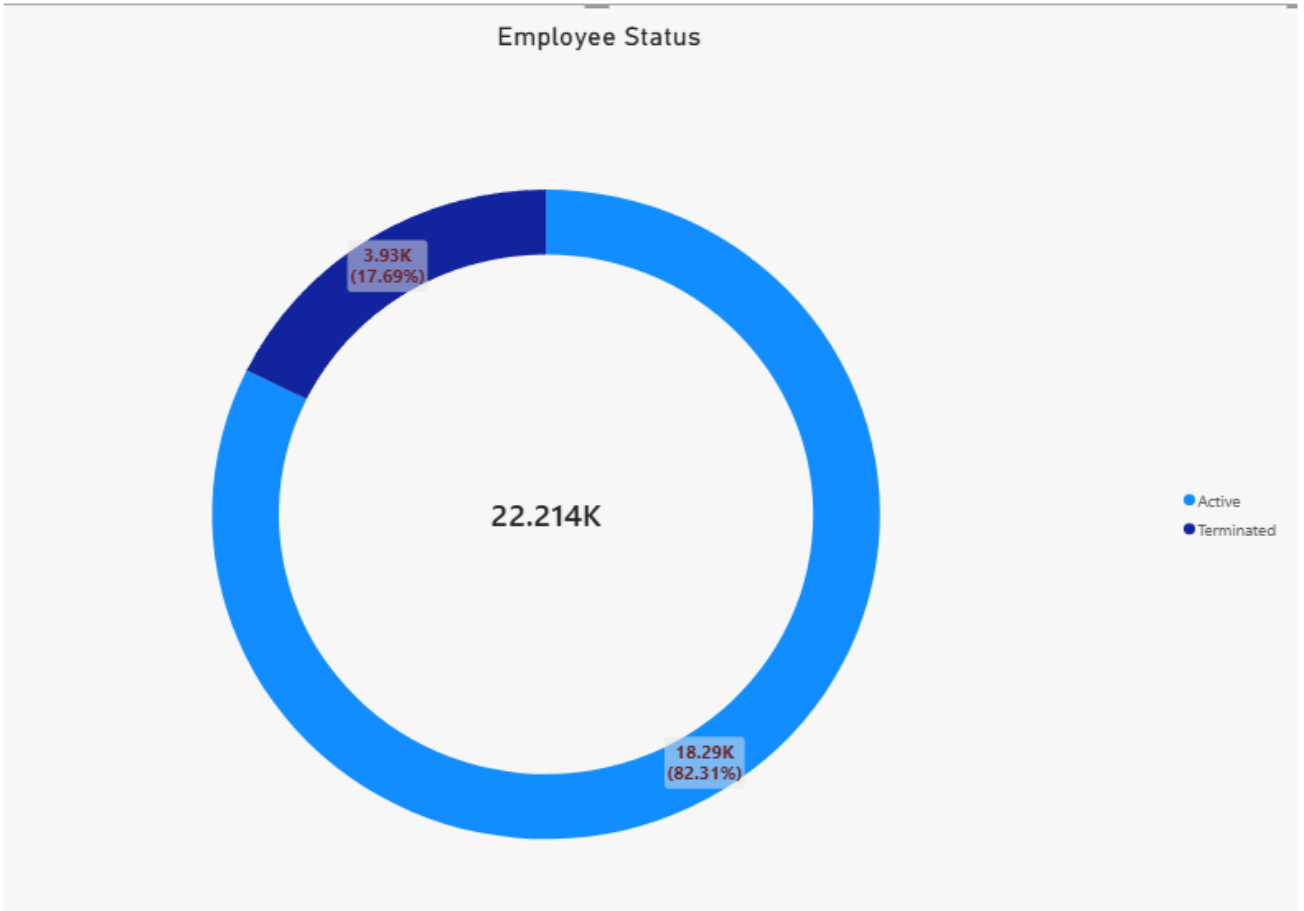


Fig 9

Shows employee distribution by age group.

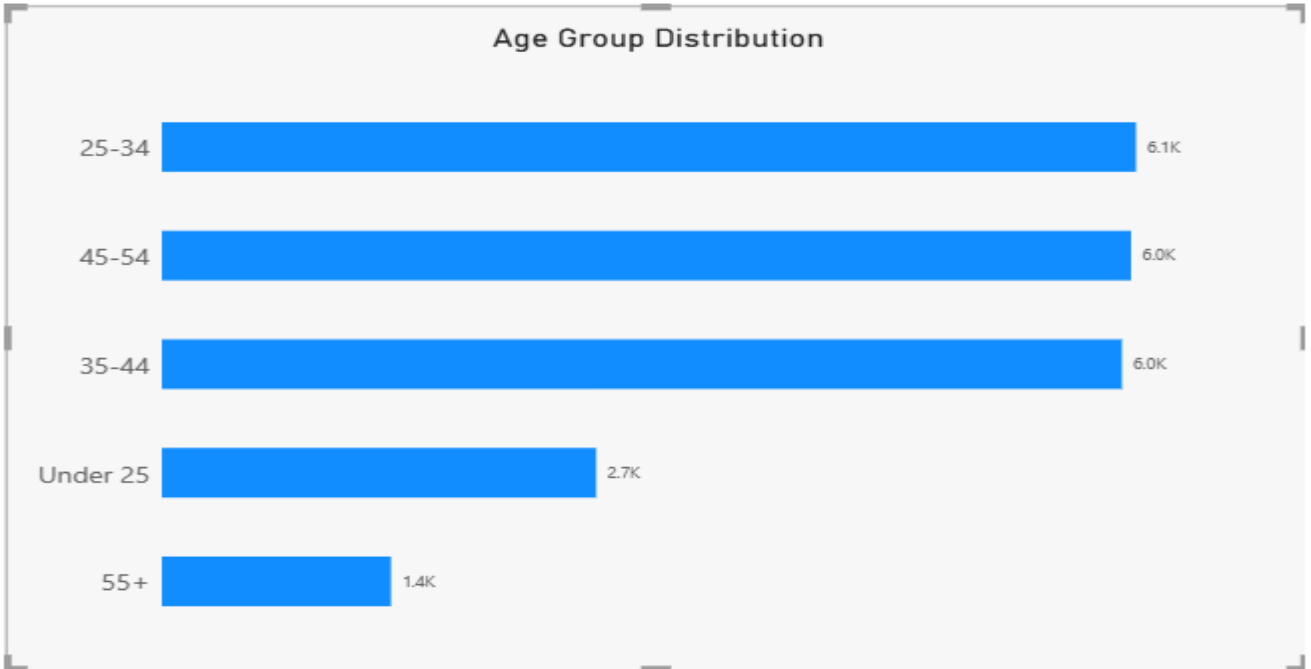


Fig 10

Compares Active and Terminated counts across departments.

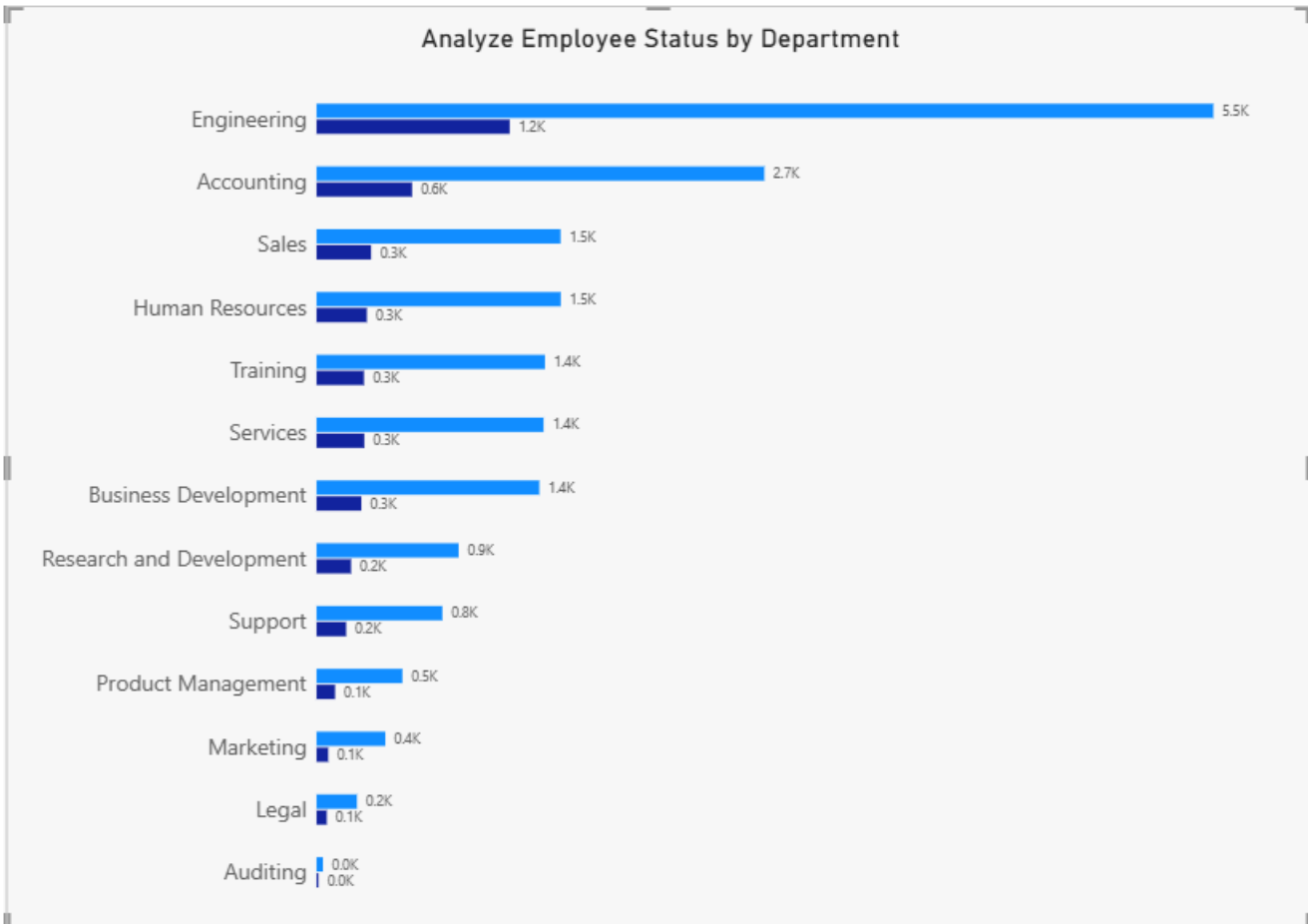


Fig 11

Chapter 7 – Key Findings

- Workforce size: 22,214 employee records are included in the analysis.
- Average age: The average employee age is 38.43 years.
- Work location: 75.25% are classified as Headquarters and 24.75% as Remote.
- State concentration: Ohio is the largest state category shown, at approximately 18.0K employees.
- Department concentration: Engineering is the largest department at approximately 6.7K employees, followed by Accounting at approximately 3.3K.
- Gender distribution: The dashboard shows approximately 50.81%, 46.46%, and 2.72% across the displayed gender categories.
- Employee status: 18.29K (82.31%) are shown as Active and 3.93K (17.69%) as Terminated.
- Age groups: 25–34, 35–44, and 45–54 are the largest groups, each around 6.0K–6.1K employees.
- Job titles: Research Assistant II has 754 employees among the displayed titles, followed by Business Analyst with 708.
- Hiring trend: Hiring counts rise sharply at the beginning of the displayed period and then remain comparatively stable with fluctuations.
- Termination trend: Termination counts rise to a visible peak and then decline toward the later years shown.

Chapter 8 – Business Suggestions

- Engineering workforce planning: Because Engineering has the largest employee population, HR can regularly review staffing needs, workload, and future skill requirements.
- Location planning: With 75.25% of employees at HQ, the company can review which roles may be suitable for remote or hybrid work based on job requirements.
- Retention investigation: Termination trends can be investigated further by department, job title, age group, and location to identify where employee exits are concentrated.
- Workforce development: The large 25–54 age groups can be considered when planning training and career-development programs.
- Geographic planning: Ohio's much larger employee concentration can be considered in workforce-capacity and hiring planning.
- Continue data-driven reporting: HR can use Power BI to monitor headcount, hiring, employee status, location, and demographic patterns over time.

These suggestions are based on descriptive dashboard patterns. They do not establish causal relationships.

Chapter 9 – Project Limitations

- The dashboard is descriptive and does not prove why employees leave or why hiring changes.
- The hiring visual represents employee counts by year rather than a formally defined hiring-rate percentage.
- Several chart labels are rounded (for example, 18.0K and 6.7K); exact underlying values should be used when precision is required.
- Employee-level HR data should be handled responsibly and only for legitimate analytical purposes.

Chapter 10 – Conclusion

The PRDA – 03 HR Analytics project demonstrates an end-to-end Power BI workflow for HR data. The data was prepared in Power Query, DAX measures were created for the main KPIs, and the results were presented through a focused set of HR visualizations. The dashboard provides a clear view of workforce size, age, work location, hiring and termination trends, department and state distribution, demographics, job titles, and employee status. These findings can support further HR investigation and workforce planning.